目录

Safety Alignment——SFT SOP

正文 1065 字（0.6 页） | 浏览 26 次（5 人） 全部 >

AI 速览 点击生成简要速览，支持更多 风格切换 助你高效阅读

https://dev.sankuai.com/code/repo-detail/~xuyiheng/mystarship/file/list

Step 1: 准备数据集

数据格式要求为 JSONL 格式，每行包含一个训练样本。示例：

```
{"messages": [{"role": "user", "content": "你好"}, {"role": "assistant", "content": "你好！有什么可以帮助你的？"}]}
```

准备好数据后，在 `starship_job_submitter/scripts/config_swift_megatron_sft.py` 的 `data_set` 内添加。示例：

代码块 Python

```python
data_set = {
    "multi_turn_v0": [
        "/mnt/dolphinfs/ssd_pool/docker/user/hadoop-xt-ai-search/ai-search/wangxiaohan17/rolling_datasets/datav0128/filtered_msgs_filtered_34k.jsonl",
        "/mnt/dolphinfs/ssd_pool/docker/user/hadoop-xt-ai-search/ai-search/wangxiaohan17/rolling_datasets/datav0128/filtered_msgs_filtered_34k.jsonl"
    ],
    "deepseekv3_distill_safe": [
        "/mnt/dolphinfs/hdd_pool/docker/user/hadoop-mtsearch-assistant-search/zhangkaizhou/safety_benchmark/eval/train/deepseekv3_sonl",
        "/mnt/dolphinfs/hdd_pool/docker/user/hadoop-mtsearch-assistant-search/zhangkaizhou/safety_benchmark/eval/train/deepseekv3_sonl"
    ],
    "both": [
        "/mnt/dolphinfs/hdd_pool/docker/user/hadoop-mtsearch-assistant-search/zhangkaizhou/safety_benchmark/dataset/split_outputs/test_chat.jsonl",
        "/mnt/dolphinfs/hdd_pool/docker/user/hadoop-mtsearch-assistant-search/zhangkaizhou/safety_benchmark/dataset/split_outputs/test_chat.jsonl"
```

```
    ],
    "positive": [
        "/mnt/dolphinfs/hdd_pool/docker/user/hadoop-mtsearch-assistan
          search/zhangkaizhou/safety_benchmark/dataset/split_outputs/t
          ive_only_chat.jsonl",
        "/mnt/dolphinfs/hdd_pool/docker/user/hadoop-mtsearch-assistan
          search/zhangkaizhou/safety_benchmark/dataset/split_outputs/t
          ive_only_chat.jsonl"
    ],
    "negative": [
        "/mnt/dolphinfs/hdd_pool/docker/user/hadoop-mtsearch-assistan
          search/zhangkaizhou/safety_benchmark/dataset/split_outputs/t
          ive_only_chat.jsonl",
        "/mnt/dolphinfs/hdd_pool/docker/user/hadoop-mtsearch-assistan
          search/zhangkaizhou/safety_benchmark/dataset/split_outputs/t
          ive_only_chat.jsonl"
    ],
    "no_system_prompt": [
        "/mnt/dolphinfs/hdd_pool/docker/user/hadoop-mtsearch-
          assistant/xuyiheng/train_wo_test_chat_no_system.jsonl",
        "/mnt/dolphinfs/hdd_pool/docker/user/hadoop-mtsearch-
          assistant/xuyiheng/train_wo_test_chat_no_system.jsonl"
    ],
    "reasoning": [
        "/mnt/dolphinfs/hdd_pool/docker/user/hadoop-mtsearch-assistan
          search/zhangkaizhou/safety_benchmark/dataset/split_outputs/b
          n_chat_with_think_answer.jsonl",
        "/mnt/dolphinfs/hdd_pool/docker/user/hadoop-mtsearch-assistan
          search/zhangkaizhou/safety_benchmark/dataset/split_outputs/b
          n_chat_with_think_answer.jsonl"
    ],
    "truncated": [
        "/mnt/dolphinfs/hdd_pool/docker/user/hadoop-mtsearch-assistan
          search/zhangkaizhou/safety_benchmark/dataset/split_outputs/m
          n_chat.jsonl",
        "/mnt/dolphinfs/hdd_pool/docker/user/hadoop-mtsearch-assistan
          search/zhangkaizhou/safety_benchmark/dataset/split_outputs/m
          n_chat.jsonl"
    ]
}
```

最后，在 `myshells/submit_32b_sft.sh` 的 `DATASETS` 中指定要用的数据集。示例：

`DATASETS=("truncated")`

Step 2: 配置训练参数

在 `starship_job_submitter/scripts/config_swift_megatron_sft.py` 中：

- 模型配置

代码块 Python

```python
# 32B 转换后的 Megatron 模型路径（需先执行 convert_32b_hf2mg.sh）
MODELS_DIR = "/mnt/dolphinfs/ssd_pool/docker/user/hadoop-ai-search/xuyiheng/models"
m_config = {
    "32B": {"tp": 4, "pp": 4,  "cp": 1, "ep": 1, "eptp": 1,  # dens ep=1
            "model_path": f"{MODELS_DIR}/Qwen3-32B-mcore",
            "model_id": f"{MODELS_DIR}/Qwen3-32B-mcore",
            "layer": 64,  # Qwen3-32B: 64 layers
            "pr": "bf16",
            "model_type": "qwen3"},
}
```

- 路径设置（保存结果）

代码块 Python

```python
PATH_PREFIX = f"/mnt/dolphinfs/ssd_pool/docker/user/hadoop-ai-search/xuyiheng/starship_result_v1"
```

- 超参数配置

代码块 Python

```python
args = [
    # 如果是huggingface的模型的话
    # '--model', m_config['model_path'],
    # '--model_type', 'qwen3_moe',
    # '--no_initialization', False,
    # 如果是megatron的模型的话
    '--load',       m_config['model_path'],

    '--dataset',    dataset[0],
    '--bf16',       True,
    #'/mnt/dolphinfs/hdd_pool/docker/share/xiaohan17/swift-train/sft_data.jsonl',
    '--save',       PATH_PREFIX + f'/model/{MODEL_DT}_{current_time
    # '--agent_template',     'qwen_zh',
    '--pipeline_model_parallel_size', m_config['pp'],
    '--tensor_model_parallel_size',   m_config['tp'],
    '--expert_model_parallel_size',   m_config['ep'],
```

```
    '--context_parallel_size',          m_config['cp'],
    '--decoder_first_pipeline_num_layers', pp_start,
    '--decoder_last_pipeline_num_layers',  pp_end,
    '--cross_entropy_loss_fusion', 'true',
    '--sequence_parallel',   True,
    '--micro_batch_size',    1,
    '--global_batch_size',   32,
    '--streaming',           False,    # for large dataset
                                       # if not using streaming, ca
                                           max_epochs, instead
                                           train/eval iters
    '--max_epochs',          1,
    '--split_dataset_ratio', 0.05,
    #'--train_iters',        255*4,    # used when not streaming
    #'--eval_iters',         5,        # used when not streaming
    '--finetune',            True,
    '--lr',                  1e-5 / (32 ** 0.5),
    '--min_lr',              5e-7,
    '--lr_warmup_iters',     10,
    '--eval_interval',       20,
    '--save_interval',       9999,
    '--max_length',          2048,
    '--tensorboard_dir',     afo_params['board.log_dir'] +
    f'/{p.dataset}/{MODEL_DT}_{current_time}',
    "--packing",             False,                    # sequence
    "--use_flash_attn",      True,
    "--recompute_granularity", "full",
    "--recompute_method",      "uniform",
    "--recompute_num_layers",  "1",

    #"--optimizer_cpu_offload", False,
    #"--optimizer_offload_fraction", "0",        # a 0-1 float or '
    #"--use_precision_aware_optimizer", "true",

    "--overlap_grad_reduce", True,
    "--overlap_param_gather", True,

    "--moe_grouped_gemm",           True,
    "--moe_shared_expert_overlap",  True,
    "--moe_aux_loss_coeff",         "5e-3",
    "--moe_router_dtype",           "fp32",
    "--moe_permute_fusion",         True,
    "--moe_shared_expert_overlap",  True,

```

```
    #"--padding_free", False,

    #"--attention_backend", "flash",
    #"--attention_backend", "fused",

    "--loss_scale",                "default",
    # extra megatron args
]
```

Step 3: 提交训练脚本

代码块 Shell

```shell
source activate swift
hope login
bash myshells/submit_32b_sft.sh
```

Step 4: 转回 hf 格式

训练后得到的 checkpoint 是 megatron 格式，需要使用脚本转为 hf 格式以供 vllm 加载推理。示例：

代码块 Shell

```shell
python3 -m scripts.convert.config_swift_convert \
  --num_gpu 8 \
  --src "/mnt/dolphinfs/ssd_pool/docker/user/hadoop-ai-search/xuyiheng/starship_result_v1/model/swift-truncated-32c-32B-truncated_02110912/ckpt/v0-20260211-091932/" \
  --dst "/mnt/dolphinfs/ssd_pool/docker/user/hadoop-ai-search/xuyiheng/models/Qwen3-32B-truncated" \
  --mg2hf \
  --h20 \
  --model_type "qwen3 --model /mnt/dolphinfs/ssd_pool/docker/user/h search/deepsearch_files_ssd/LLMbasemodels/swift_mcore/Qwen/Qwen3-
```

仅供内部使用，未经授权，切勿外传